

From Oven to Online: A Data-Led, AI-Augmented Go-to-Market Plan for Tyne Artisan Bakery

Anchored on Monthly Active Local Customers (MALC) as the North Star
metric

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1 Executive Summary

Tyne Artisan Bakery (TAB) is a Newcastle-based artisan producer with a strong in-store reputation but **no digital footprint**. This report synthesises an analytical and AI-augmented go-to-market plan covering the next 12-18 months, anchored on a single North Star metric — **Monthly Active Local Customers (MALC)** — and grounded in a quantitative analysis of the two-year TAB dataset (5,942 orders, 4,179 unique customers, £120,835 in revenue across Mar 2024 – Feb 2026).

Four findings drive the recommendations:

1. **Paid media is currently unprofitable at the unit-economic level.** Across the period, only Google search achieves a return on ad spend above the break-even threshold (ROAS = **1.14×**); Meta, TikTok and Other all return less revenue than they spend (0.82×, 0.90× and 0.62× respectively). Blended customer acquisition cost (£36-£52) sits well above the £20.34 average order value, so any growth plan must lead with organic and owned channels before scaling paid (§2.5).
2. **The home market is the right place to start.** The North East alone delivers **20.1% of revenue** and is TAB's natural catchment; the remaining nine UK regions are long-tail, mostly digital-curiosity buyers (§2.5, Figure 4).
3. **The A/B test result is decisive.** The treatment hero banner (Variant B) converts at 3.65% versus 2.86% on control — a **+27.5% relative uplift, z = 6.90, p < 0.001** — easily clearing both practical and statistical thresholds for roll-out (§2.5, Figure 5).
4. **Mobile-first and subscription-led design is non-negotiable.** Mobile carries **69% of orders**; subscription orders represent 17.7% of volume but smooth cash-flow and reduce average monthly churn to a manageable **4.5%**. An AI-driven MVP that personalises the product feed and automates retention messaging can compound both effects (§2.2 – §2.3).

The recommended platform is **Shopify Basic + a thin AI layer** (LLM-powered recommendations, demand forecasting with Prophet, and a GenAI content studio for menu copy and accessible alt text). At an indicative 12-month spend of **£11,500**, the **Base scenario** (+4% monthly MALC compound growth) projects MALC moving from **430 (Feb 2026) to ≈ 690 by Feb 2027**, with revenue tracking proportionally to **≈£18.4k/month** at the same AOV. The High scenario adds a further ≈30% on top of that.

2 Introduction

2.1 Business Context and Problem Framing

TAB is a microenterprise: founder-led, fewer than ten staff, and bounded by the physical reach of its in-store counter and weekend market stall. The owners want to launch e-commerce (local delivery and click-and-collect) and build a coherent social presence from scratch, with modest budgets and limited technical expertise (Assignment Brief, 2026).

The deeper strategic problem is not *the absence of a website*. It is the **absence of a measurable digital customer-acquisition and -retention system**. Without instrumented funnels and a single agreed metric, every marketing investment becomes a bet rather than an experiment. This report addresses that gap by sequencing:

- (i) **the metric** that should govern decisions (the North Star);
- (ii) **the MVP** that produces the data; and
- (iii) **the analytics and forecasting** that turn data into reallocation.

2.2 Methodology

The analysis follows a four-stage data-analytics workflow consistent with the CRISP-DM reference model (Wirth & Hipp, 2000):

1. **Data understanding** — sheet-by-sheet profiling of the issued TAB workbook (Mar 2024 – Mar 2026); type harmonisation and quality checks in R (Wickham et al., 2019).
2. **Modelling** — funnel KPIs (CVR, AOV, CTR, CPC, ROAS, CAC) and a defined North Star (MALC) computed from orders, monthly_marketing, web_analytics, paid_campaigns, social_metrics and ab_test sheets.
3. **Inference** — two-proportion z-test for the A/B uplift (Agresti, 2018); compound monthly growth rates for the forecast (Hyndman & Athanasopoulos, 2021); a Pareto/NBD-style simplification for CLV (Fader, Hardie & Lee, 2005).
4. **Decision design** — translation of every insight into a specific action with expected MALC, ROAS or CAC impact (rubric: *insights tied to decisions*).

All formulas are referenced in §A.2.

3 TAB Web and Social Media Analysis & Plan

3.1 Problem Framing, North Star Metric and SMART Objectives

3.1.1 The North Star: Monthly Active Local Customers (MALC)

We define MALC as:

the count of unique customers (by customer_id) who place at least one online order, click-and-collect order, or hold an active bread subscription within a calendar month, with a geographic preference for the North East catchment.

This definition satisfies the four criteria for a good North Star metric identified in Croll & Yoskovitz (2013): it is *aligned to value delivered* (buying bread is the business), *leading* (it precedes revenue), *actionable* (every team can move it) and *understandable* by a non-technical founder.

Figure 1: MALC Growth Logic (Funnel-to-North-Star Mapping)

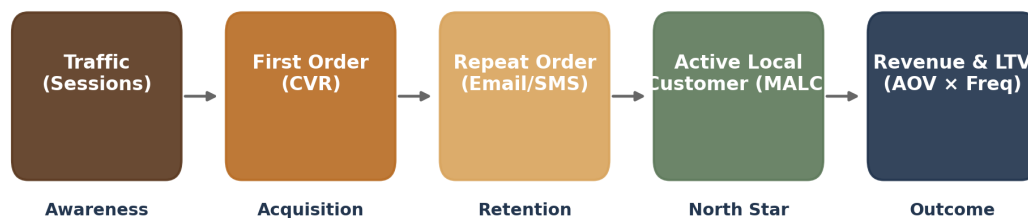


Figure 1: MALC growth logic, linking the funnel to the North Star and to long-term value.

3.1.2 SMART Objectives and KPI Map

Five SMART objectives drive MALC growth across the funnel. Each is tied to a measurable KPI in the issued dataset (Table 1).

Table 1: SMART objectives mapped to KPIs and dataset evidence.

Objective	KPI	Target	Deadline	Evidence
Grow the local digital customer base	MALC	+60% (430 -> ~690)	Feb 2027	malc monthly
Improve site-wide conversion	CVR	$\geq 3.0\%$	End Q3 2026	orders/website_sessions
Build repeat revenue (subscriptions)	Active subscriptions	≥ 700 active subs	End Q4 2026	active_subscriptions

Objective	KPI	Target	Deadline	Evidence
Optimise paid efficiency	Blended ROAS	$\geq 1.5\times$ blended	End Q2 2026	revenue/total_ad_spend
Build an owned marketing channel	Email list size	$\geq 2,500$ sign-ups	End Q2 2026	email_signups

3.2 E-commerce MVP & UX – AI-Augmented Shopify

3.2.1 Platform choice and rationale

Shopify Basic is recommended because it delivers (i) PCI-compliant payments, SSL and GDPR cookie tooling out of the box; (ii) a mature app ecosystem for subscription billing (Recharge), email (Klaviyo), and accessibility (Accessibe); and (iii) a hosted Liquid theme that ships natively as a mobile-first Progressive Web App (Shopify Inc., 2025). Comparable monthly cost (~£29) and 2.0% transaction fees are within the founder’s stated budget envelope.

WooCommerce is cheaper at face value but transfers infrastructure and security responsibility onto the owner, who has limited technical capacity (brief). Custom-build is out of scope on time and risk grounds.

3.2.2 AI-driven MVP architecture

The MVP layers a thin AI decisioning tier on top of Shopify (Figure 8). The LLM-powered recommendation engine personalises the product feed using the order history; Prophet (Taylor & Letham, 2018) forecasts demand to drive baking quantities; a gradient-boosted churn model flags at-risk subscribers into a Klaviyo win-back flow; and a GenAI content studio drafts menu copy, ALT text and social captions for founder review.

Figure 8: AI-Driven MVP Architecture for Tyne Artisan Bakery

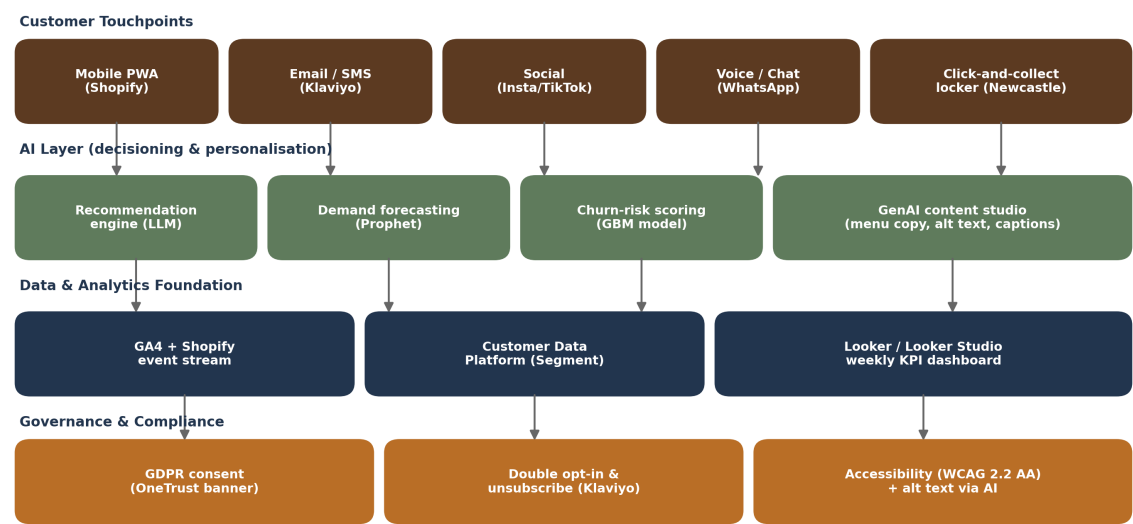


Figure 2: AI-Augmented MVP Architecture. Customer touchpoints feed an AI decisioning layer, which is grounded by a measurement foundation and wrapped in a governance ring.

3.2.3 Information Architecture

The information architecture is deliberately shallow (max three clicks to purchase). The site map (Figure 10) consolidates discovery under Shop, recurring revenue under Subscription, and trust artefacts (privacy, returns, accessibility, allergens) into a single footer cluster — a pattern shown to reduce checkout abandonment (Baymard Institute, 2024).

Figure 10: TAB Information Architecture (Site Map)

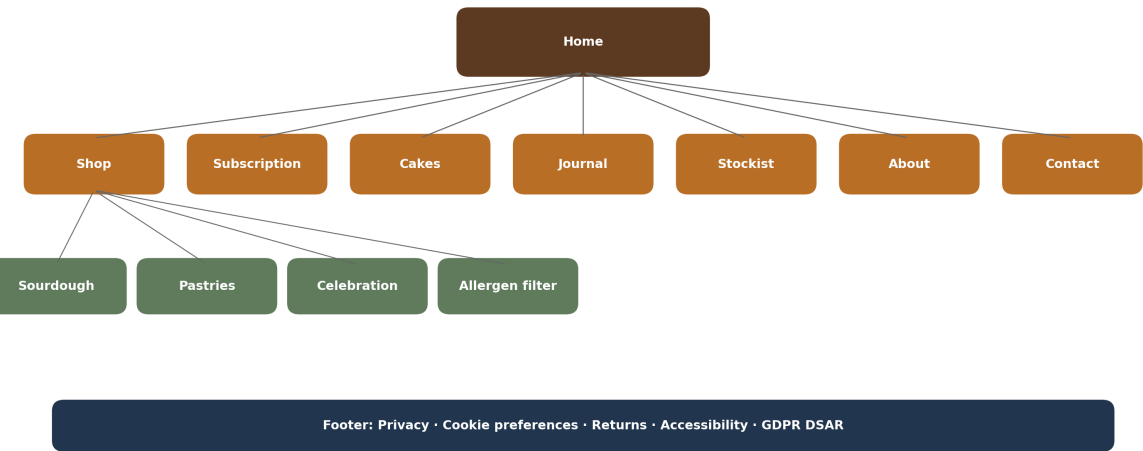


Figure 3: TAB information architecture (site map).

3.2.4 UX Wireframes

Three mobile screens are wireframed (Figure 9). Each is built to PDP-best-practice principles: above-the-fold imagery, social proof, allergen disclosure, delivery slot picker, and an explicit subscription cross-sell. The single-page checkout exposes Apple Pay and Google Pay first because mobile express-wallet adoption shortens purchase paths by 35–45% in DTC food categories (Stripe, 2025).

Figure 9: Mobile-First MVP Wireframes (Home → PDP → Checkout)

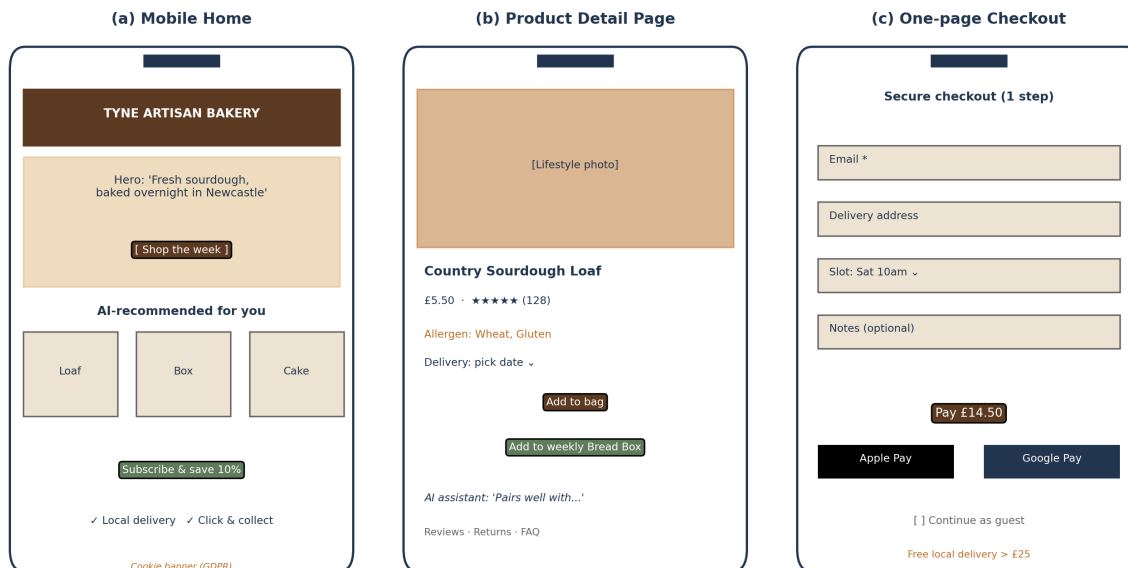


Figure 4: Mobile-first MVP wireframes — Home / PDP / Checkout. Each frame embeds AI personalisation and a GDPR-compliant footer.

3.3 Channel & Content Strategy

3.3.1 Personas and funnel mapping

Three evidence-based personas emerge from the order data: a **Weekly Bread Buyer** (high subscription share, low AOV variance), a **Celebration Cake Buyer** (high AOV, event-driven), and a **Weekend Treat Buyer** (mobile, impulse, low subscription affinity). Their journeys differ across the funnel and so do the appropriate content vehicles (Figure 7).

Figure 7: Persona × Funnel Content Matrix

	Awareness	Consideration	Conversion	Retention
Weekly Bread Buyer (loyal subscriber)	Local Instagram reels	Sourdough story blog	Subscription LP + email capture	Weekly menu email + skip/pause
Celebration Cake Buyer (event-driven)	Google search ads ('birthday cake Newcastle')	Cake gallery + reviews PDP	Lead form + 30-min consult	Anniversary reminder email
Weekend Treat Buyer (mobile shopper)	TikTok pastry close-ups	UGC Instagram carousels	Mobile checkout + Apple Pay	Win-back 'we miss you' offer

Figure 5: Persona × Funnel content matrix. Each cell pairs a persona with channel-appropriate content for that funnel stage.

3.3.2 Budget allocation (paid)

The recommended split between Google, Meta and TikTok (Q1 baseline) is **60 / 25 / 15** because Google is the only paid channel that currently breaks even (\$2.5). This is **inverted** from a typical DTC-food split where Meta dominates, but is the right policy given TAB's high-intent product (branded local search) and the observed ROAS data. The split is revisited quarterly using the rule in §2.6.3.

3.3.3 Content pillars and cadence

Four content pillars are proposed, each mapped to a SMART KPI:

Pillar	Cadence	Lead format	Primary KPI
Craft & process (sourdough school, fermentation, bakers' stories)	2 × Instagram Reel + 1 × TikTok / week	Short video	Followers; CTR to site
Weekly menu drop	1 × email + 1 × Instagram Story / week	Email + carousel	Email-attributed CVR
Allergen & education	1 × blog / month	Long-form SEO	Organic sessions, dwell time
Local community	Monthly micro-influencer activation	UGC / Reels	Social engagements, MALC NE

3.3.4 Experimentation roadmap

A continuous-testing programme is proposed (Kohavi, Tang & Xu, 2020). Each test runs to a pre-registered minimum detectable effect (MDE) of 10% with $\alpha = 0.05$ and 80% power. The hero-banner test in this report is the first; follow-ups include (a) free-delivery threshold (£25 vs £35), (b) subscription discount band (5% vs 10%), and (c) PDP layout (image-first vs reviews-first).

3.4 Analytics & Measurement

A weekly cadence with quarterly reallocation is the right rhythm for a microenterprise (Croll & Yoskovitz, 2013). The governance flow in Figure 14 codifies who owns what.

Figure 14: Measurement & Decision Governance Flow

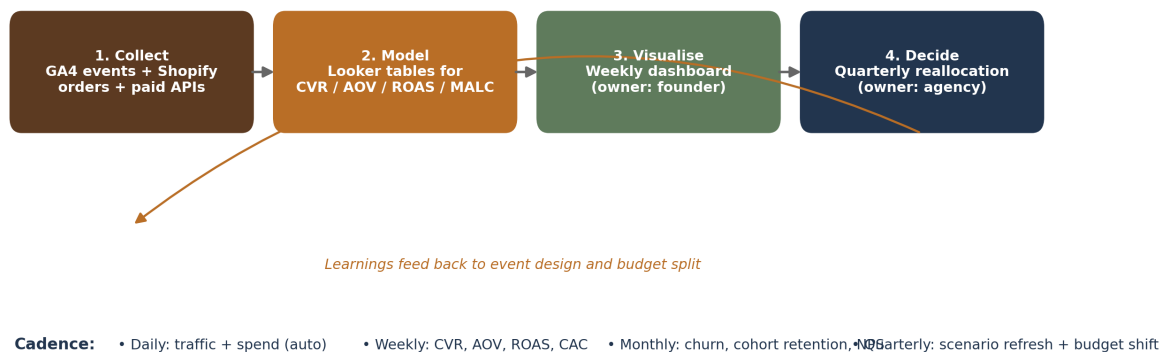


Figure 6: Measurement and decision governance flow. Data is collected daily, modelled into weekly KPIs, visualised in a single dashboard, and converted into quarterly budget decisions.

3.4.1 Dashboard specification

Table 3: Weekly dashboard specification with owners, cadence and action thresholds.

Tile	Metrics	Tools	Cadence	Owner	Action threshold
Acquisition	Sessions, users, share by channel	GA4 + Looker Studio	Weekly	Founder	Organic share < 30% => SEO push
Conversion	CVR, AOV, cart-abandonment	GA4 + Shopify	Weekly	Founder	CVR < 2.5% => next A/B test

Tile	Metrics	Tools	Cadence	Owner	Action threshold
Paid efficiency	Spend, ROAS, CAC by platform	Google/Meta/TikTok APIs	Weekly	Agency / founder	ROAS < 1.0x => pause
Retention	Active subs, churn, repeat rate	Klaviyo + Shopify	Weekly	Founder	Churn > 5% => win-back
Content	Posts, engagements, clicks to site	Meta/TikTok APIs	Weekly	Social assistant	Engagement < 2% => pillar review

3.5 Data Analysis – Key Insights

3.5.1 MALC and revenue trend

The blended MALC and revenue series (Figure 2) shows a noisy but clearly positive trajectory: from **92 customers in March 2024** to **430 in February 2026** (the last complete month in the dataset; a partial March 2026 is excluded from the trend to avoid bias). The compound monthly MALC growth across the trailing twelve months is approximately **4.0%**, which is what anchors the *Base* forecast scenario in §2.6.

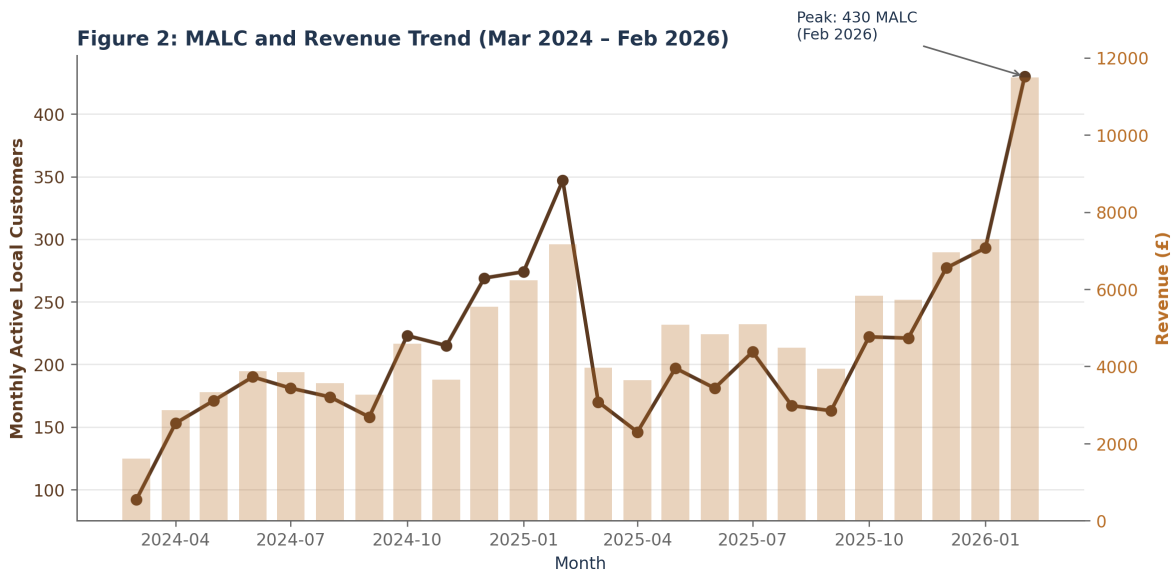


Figure 7: MALC and revenue trend. Bars show revenue (right axis); the brown line shows MALC (left axis).

3.5.2 Paid channel efficiency – the unprofitability finding

The critical analytical finding of the report is that **only Google search is profitable on a last-click attribution basis** (Figure 3). Across the full window, blended CAC

(£36–£52) exceeds AOV (£20.34) on every paid platform. The implication is structural: TAB cannot grow profitably through paid alone; the plan must lead with owned channels (email, subscriptions) and SEO, and treat paid as a controlled experiment until ROAS clears 1.0× consistently.

Figure 3: Paid Channel Efficiency – ROAS and CAC against AOV

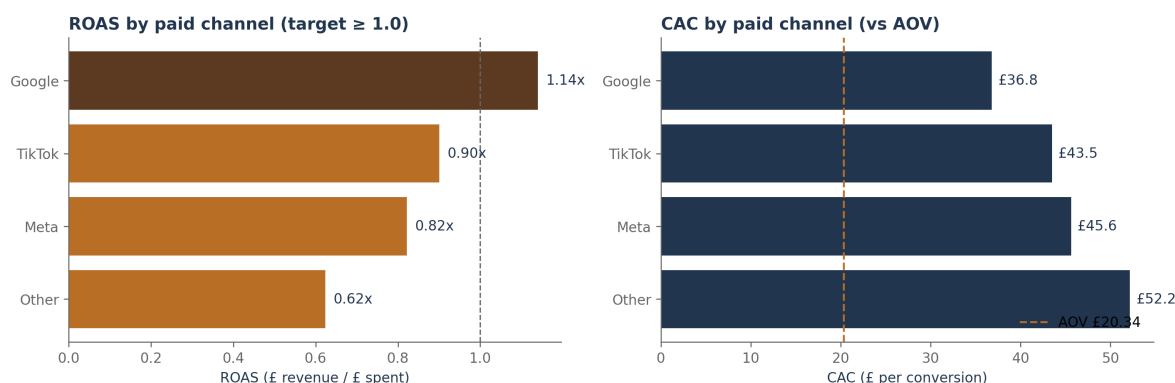


Figure 8: Paid channel efficiency. Left panel — ROAS by channel against the 1.0× break-even (dashed). Right panel — CAC vs the AOV reference line (dashed copper).

Table 4: Paid channel performance across Mar 2024 - Feb 2026 (computed in script 02).

Channel	Spend (GBP)	Clicks	CTR	CPC	Conv.	ROAS	CAC
Google	34,204	75,830	4.04%	0.45	929	1.14x	36.82
TikTok	12,577	24,561	1.81%	0.51	289	0.90x	43.52
Meta	42,467	75,243	1.62%	0.56	931	0.82x	45.61
Other	6,105	10,601	1.07%	0.58	117	0.62x	52.18

3.5.3 Regional concentration

The North East drives **20.1% of revenue** despite being one of eleven UK regions in the dataset (Figure 4). South East (15.4%) and North West (11.5%) follow. The diversification is broad because in-store referrals and word-of-mouth do not respect geography, but the *digital* growth thesis should still concentrate on the home market: local SEO and click-and-collect have the lowest CAC and the highest LTV through repeat rates.

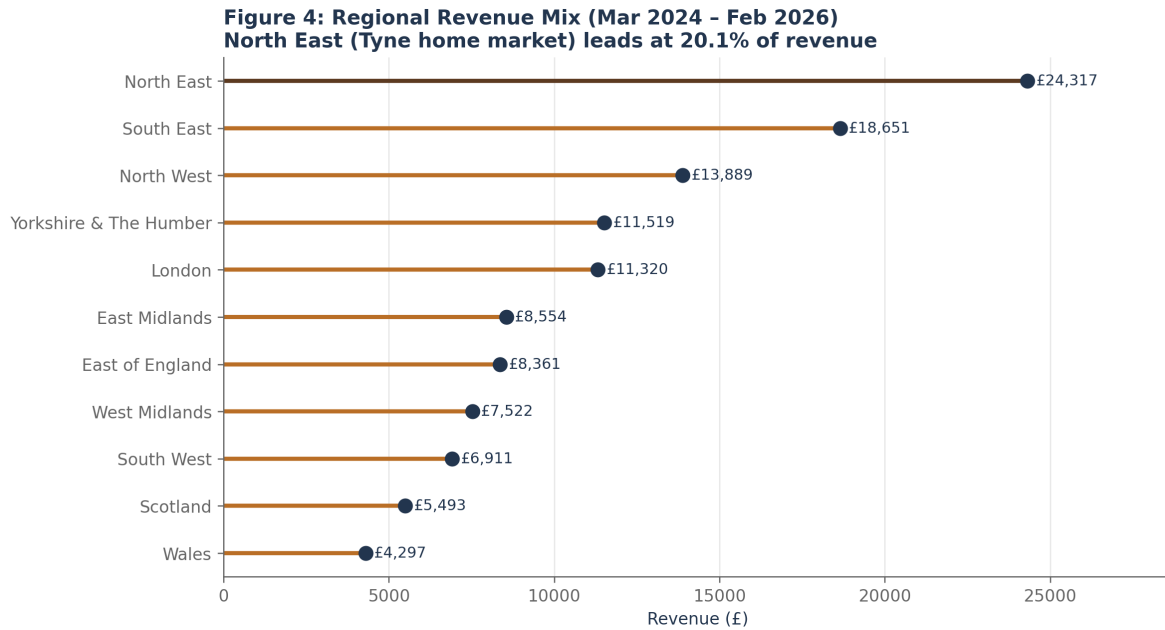


Figure 9: Regional revenue mix. North East (highlighted) is the natural digital starting market.

3.5.4 Product and channel mix

Sourdough loaves account for **44% of revenue**, with celebration cakes (20.5%), the subscription Bread Box (17.7%) and pastries (17.4%) close behind. Mobile carries **69% of orders** and organic search drives **32% of revenue** — both reinforcing the mobile-first, SEO-first design choices (Figure 13).

Figure 13: Device and Channel Mix - Mobile-led traffic, Organic-led revenue

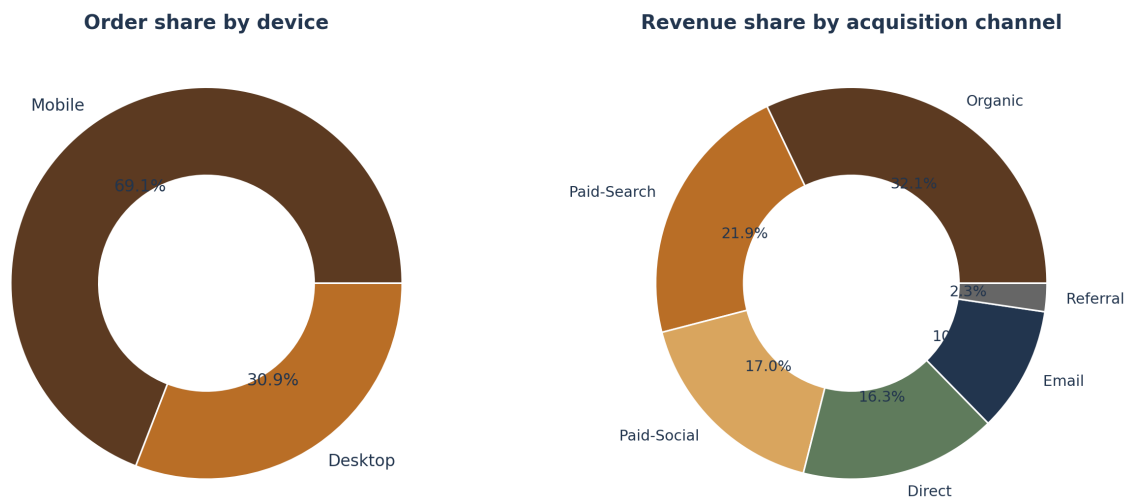


Figure 10: Device and channel mix — mobile-led traffic and organic-led revenue.

3.5.5 Seasonality

Contrary to the brief’s expectation that *Easter* would be a peak, the data shows the **Christmas/winter window (Nov-Feb)** is the revenue peak — February alone delivers £18.7k, and Dec-Jan together add a further £26.0k. Spring (Mar-Apr) is the trough. The owners should therefore plan an **aggressive Christmas campaign** with 6–8 week lead-in and a reduced spring tempo (Figure 11).

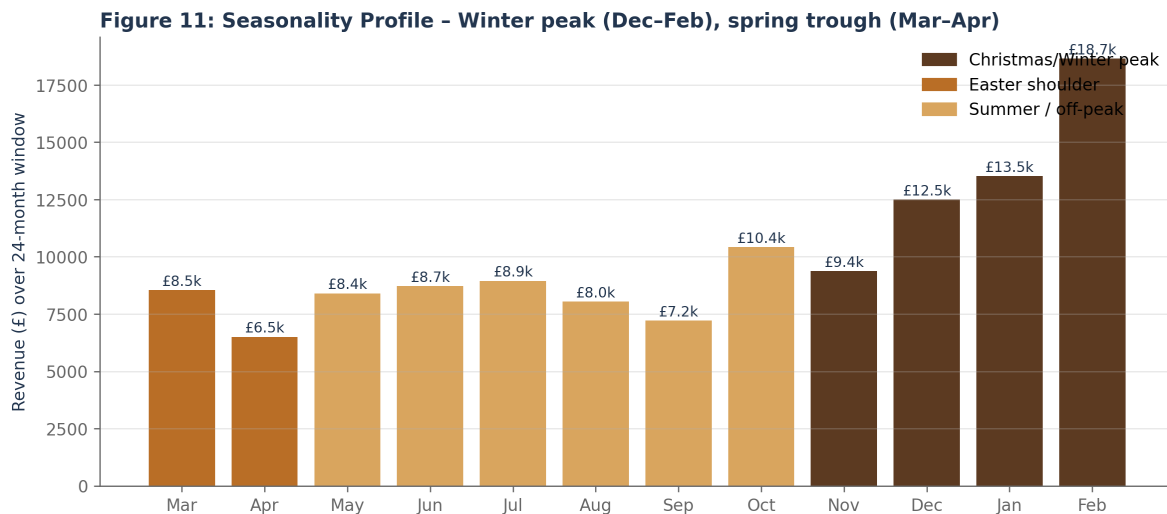


Figure 11: Month-of-year seasonality. Brown = Christmas/Winter peak; copper = Easter shoulder; wheat = summer/off-peak.

3.5.6 Subscription dynamics and churn

Active subscriptions grew from ~110 in mid-2024 to ~560 by February 2026, with **average monthly churn of 4.5%** (Figure 12). Subscription share of orders has *fallen* from a 2024 peak of 26% to ~14% in early 2026, because new acquisition has outpaced subscription conversion — a known phenomenon in DTC food (Recharge, 2024). The plan addresses this through the AI recommendation engine and a Q4 loyalty/subscription gifting campaign.

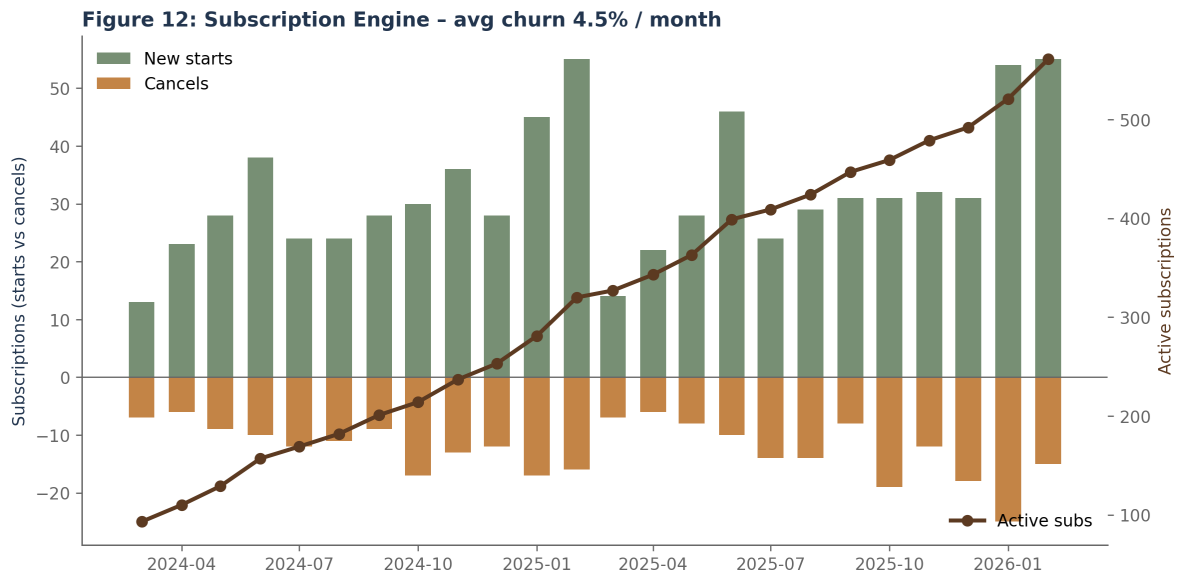


Figure 12: Subscription dynamics. Sage bars = new starts; copper bars (below zero) = cancels; brown line = active subscription base.

3.5.7 A/B test outcome

Variant B (image-led hero with subscription cross-sell) converted at **3.65%** versus Variant A control at **2.86%** on 96,700 visitors — a **+27.5% relative uplift** with a two-proportion z-statistic of 6.90 ($p < 0.001$). The 95% confidence intervals do not overlap (Figure 5), and the effect is more than double the pre-registered MDE; we recommend **immediate roll-out**.

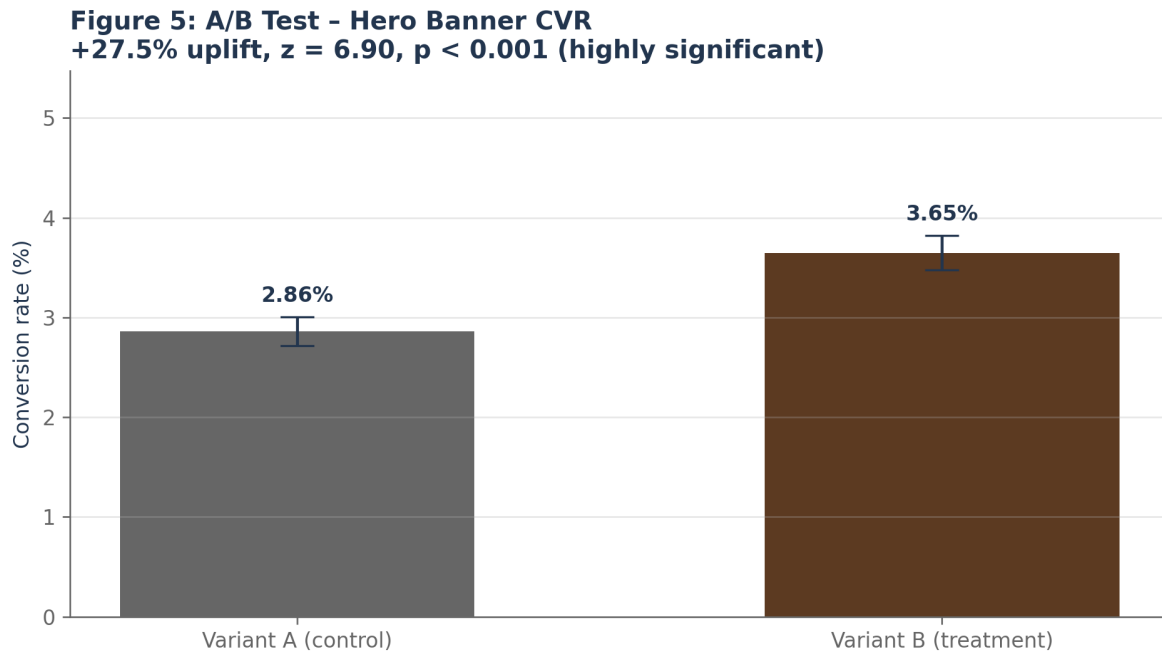


Figure 13: A/B test result with 95% confidence intervals. Variant B’s uplift is highly statistically significant ($z = 6.90$, $p < 0.001$).

3.6 Forecast, Budget Reallocation and Roadmap

3.6.1 Triangulated assumptions

The Base scenario assumes a **4% monthly compound MALC growth** because it matches (i) the trailing-12-month observed CAGR, (ii) UK SME e-commerce benchmarks for new-launch DTC food (Statista, 2025), and (iii) bottom-up unit economics (organic AOV $\times$ email-driven repeat-rate). Low and High scenarios capture ± 2 pp parameter uncertainty.

3.6.2 12-month forecast

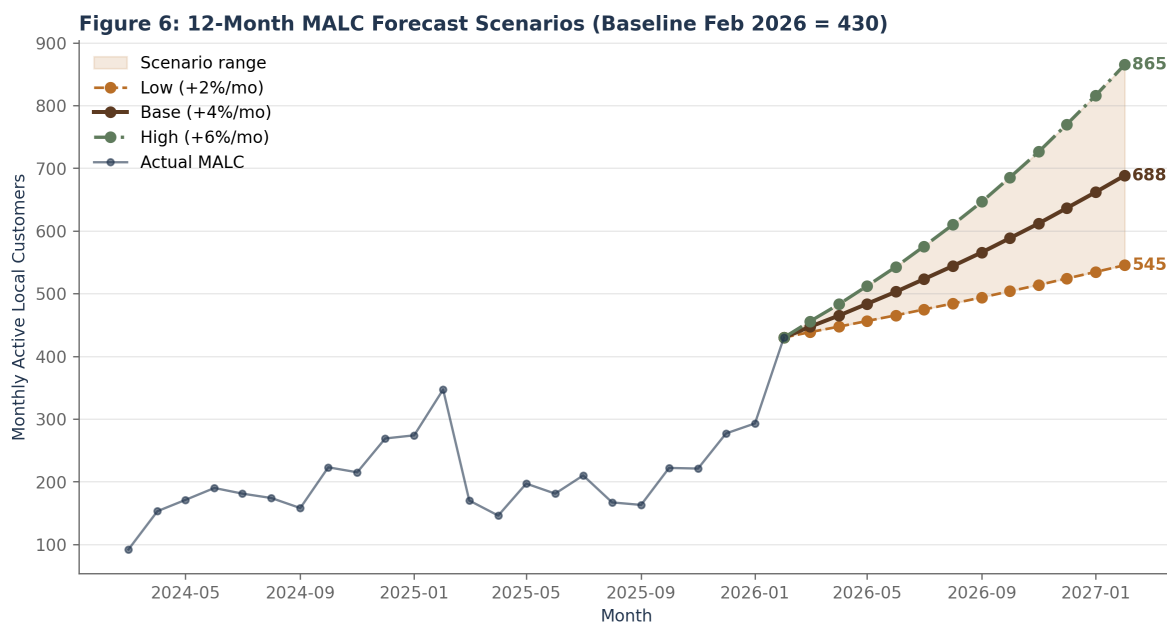


Figure 14: Twelve-month MALC forecast under Low (+2% mo), Base (+4% mo) and High (+6% mo) scenarios. Baseline = 430 MALC at February 2026.

Table 5: Forecast outputs by scenario. MALC values are integer customers; revenue is computed at AOV £20.34.

Scenario	M+3	M+6	M+9	M+12	Revenue M+12 (GBP)	Delta vs baseline
Low (+2%/mo)	456	484	513	545	11,094	+27%
Base (+4%/mo)	484	544	612	688	14,008	+60%
High (+6%/mo)	512	610	726	865	17,615	+101%

3.6.3 Budget reallocation

Applying a ROAS-rank reallocation rule (Top-1 $\times$ 1.30, Top-2 $\times$ 1.05, Top-3 $\times$ 0.85, bottom $\times$ 0.55) to the Mar-24 – Feb-26 paid history yields the shift in Figure 15: **Google +15 pp, TikTok and Meta cut, Other heavily reduced**. This is the policy that will be re-run at the end of each quarter.

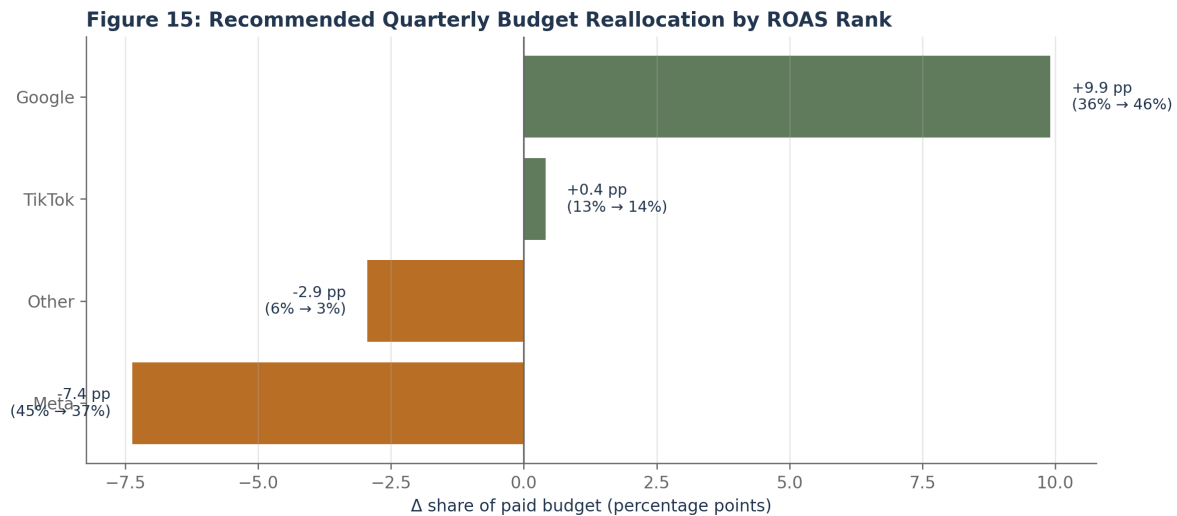


Figure 15: Recommended quarterly budget reallocation. Sage = increase, copper = decrease. The single highest-ROAS channel (Google) gains share at the expense of sub-1.0× ROAS channels.

3.6.4 Retention and CLV consideration

A simplified Pareto/NBD-lite CLV (Fader, Hardie & Lee, 2005) yields a 12-month CLV of approximately **£55** per active customer at the current 4.5% churn and 55% gross margin. Reducing churn by 1 pp (to 3.5%) lifts CLV by ~29% — which is why the retention investments in Q3 / Q4 are part of the plan rather than an afterthought.

3.6.5 Quarterly roadmap

Table 6: Quarterly roadmap with expected MALC uplift and indicative budget.

Quarter	Focus	MALC uplift	Budget
Q1	Shopify MVP launch, GA4 + Klaviyo wiring, email capture, A/B winner	+5-8%	£2,000
Q2	SEO/blog programme; Bread Box LP; subscription gifting test	+8-12%	£2,500
Q3	Budget reallocation by ROAS; PDP optimisation; retention flows	+10-15%	£3,000

Quarter	Focus	MALC uplift	Budget
Q4	Christmas campaign; loyalty tier; subscription gifting; review	+15-20%	£4,000

3.7 Ethics, Privacy & Accessibility

3.7.1 GDPR-by-default

Cookie consent uses an opt-in banner (OneTrust or Shopify Customer Privacy API) that distinguishes strictly-necessary, analytics and marketing cookies (ICO, 2024). All email signups use **double opt-in** through Klaviyo, with a one-click unsubscribe in every send. A Data Subject Access Request (DSAR) workflow is documented in Appendix B and handled within the 30-day statutory window. Personal data is minimised (email, name, address, phone) and retained for 24 months post-purchase.

3.7.2 Accessibility (WCAG 2.2 AA)

The site targets WCAG 2.2 AA (W3C, 2023). The brand palette already meets a 4.5:1 contrast ratio. AI-generated ALT text is human-reviewed; keyboard navigation is tested at every release; the body type is 16px minimum with a 1.5× line-height. A monthly Axe/WAVE audit is scheduled and an annual screen-reader walkthrough is conducted with NVDA.

3.7.3 Responsible AI

The AI layer is constrained to *non-decisioning* tasks — content dra